WORKED SOLUTION

STEP III 1994, Question 7

Pure • Group Isomorphisms

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q7

The key idea

Two groups can only be isomorphic if they share every structural property that an isomorphism preserves. The two most useful such properties are the list of element orders and whether the group is commutative (abelian). Listing the orders already separates S_3 from Z_6. To prove that two groups *are* isomorphic, we write down an explicit bijection and check that it respects the operations. The final part comes down to commutativity: every group made of complex numbers is abelian, and S_3 isn't.

Solution

The elements of S_3 and Z_6

Label the three objects $1, 2, 3$ and use cycle notation, composing from right to left, so $\sigma\tau$ means "do τ first, then σ". The six permutations are listed below.

element of S_3	e	(12)	(13)	(23)	(123)	(132)
order	1	2	2	2	3	3

A transposition swaps two objects, and doing it twice restores everything, so it has order 2. A 3-cycle such as (123) sends $1 \to 2 \to 3 \to 1$, so it needs three applications to return each object home, and it has order 3.

In $Z_6 = \{0, 1, 2, 3, 4, 5\}$ the identity is 0, and the order of a is the least $n \geq 1$ with $na \equiv 0 \pmod 6$, which is $6/\gcd(a, 6)$.

element of Z_6	0	1	2	3	4	5
order	1	6	3	2	3	6

For example, $4 + 4 = 8 \equiv 2$ and $4 + 4 + 4 = 12 \equiv 0$, so 4 has order 3.

Are S_3 and Z_6 isomorphic?

No. An isomorphism θ preserves orders: if g has order n then $\theta(g)^k = \theta(g^k)$, and this is the identity exactly when g^k is, because θ is injective. So isomorphic groups have the same number of elements of each order. But Z_6 has two elements of order 6 and S_3 has none, so they aren't isomorphic. (They also differ in their elements of order 2: S_3 has three of them and Z_6 has only one.)

There is a second reason, which we'll reuse at the end. Z_6 is abelian, since addition is commutative. S_3 isn't:

$$(12)(13) = (132), \qquad (13)(12) = (123).$$

For the first product, (13) sends $1 \to 3$ and then (12) fixes 3, so $1 \to 3$. Similarly $3 \to 2$ and $2 \to 1$. An isomorphism carries commuting pairs to commuting pairs, so a non-abelian group can't be isomorphic to an abelian one.

C_6 is isomorphic with Z_6

Define $\phi : Z_6 \to C_6$ by $\phi(k) = \alpha^k$ for $k = 0, 1, \ldots, 5$.

- *Bijective.* The numbers $\alpha^k = e^{ik\pi/3}$ for $k = 0, \ldots, 5$ have arguments $0, \frac{\pi}{3}, \ldots, \frac{5\pi}{3}$. These are distinct angles in $[0, 2\pi)$, so the six values are distinct points on the unit circle. Hence ϕ is injective, and so it is a bijection between two sets of size 6.
- *Operations.* Let $a, b \in Z_6$, and write $a + b = 6q + r$ with $r = (a + b) \bmod 6$. Since $\alpha^6 = e^{2\pi i} = 1$,

$$\phi(a)\,\phi(b) = \alpha^a \alpha^b = \alpha^{a+b} = (\alpha^6)^q \alpha^r = \alpha^r = \phi\big((a + b) \bmod 6\big).$$

So ϕ turns addition mod 6 into complex multiplication, and it is an isomorphism. Hence $C_6 \cong Z_6$. As a sanity check, α and α^5 (the images of 1 and 5) have order 6, α^2 and α^4 have order 3, and $\alpha^3 = -1$ has order 2, which matches the table for Z_6.

A subgroup of $\mathbb{C}$ isomorphic with S_3?

> **Ask yourself** What property do *all* groups of complex numbers share?

They are all abelian. A subgroup H of $(\mathbb{C}, +)$ is abelian because $z + w = w + z$ for all complex z, w. A subgroup H of the non-zero complex numbers under multiplication is abelian because $zw = wz$. A subgroup uses the same operation as the big group, so it inherits commutativity.

Now suppose some such H had an isomorphism $\theta : S_3 \to H$. Apply it to the non-commuting pair above. Since H is abelian,

$$\theta\big((12)(13)\big) = \theta(12)\,\theta(13) = \theta(13)\,\theta(12) = \theta\big((13)(12)\big),$$

written multiplicatively (replace the products by sums for the additive case). Because θ is injective, this would give $(12)(13) = (13)(12)$, which is false. So **no** subgroup of the complex numbers, additive or multiplicative, is isomorphic with S_3.

Checks

- In each group the orders divide 6 (Lagrange's theorem): $1, 2, 3$ in S_3 and $1, 2, 3, 6$ in Z_6.
- A computer recomputed every order in S_3, Z_6 and C_6, confirmed $(12)(13) \neq (13)(12)$, and checked $\alpha^{(a+b) \bmod 6} = \alpha^a \alpha^b$ numerically for all 36 pairs (a, b), to within 10^{-12}.

Common slips

- Saying "both have six elements, so they're isomorphic". Having the same size is necessary but nowhere near sufficient.
- Saying the elements of order 3 in S_3 are (12) and (13). Transpositions have order 2; only the 3-cycles have order 3.
- In the last part, only ruling out *finite* multiplicative subgroups (the roots of unity). The commutativity argument covers every subgroup, finite or infinite, additive or multiplicative, at once.

Beyond STEP

This question is about invariants: properties, such as element orders and commutativity, that isomorphic groups must share. In fact every group of order 6 is isomorphic to either Z_6 or S_3. More generally, any finite subgroup of the non-zero complex numbers under multiplication is cyclic, so it is a group of roots of unity. You'll meet both results early in a first course on groups. Allenby's *Rings, Fields and Groups* is a friendly place to read about them.

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